

Week 3

- Bayes Theorem
- χ^2 Test of Independence

χ^2 -Test of Independence:

H_0 : Variables are Independent

H_a : Var. are NOT independent

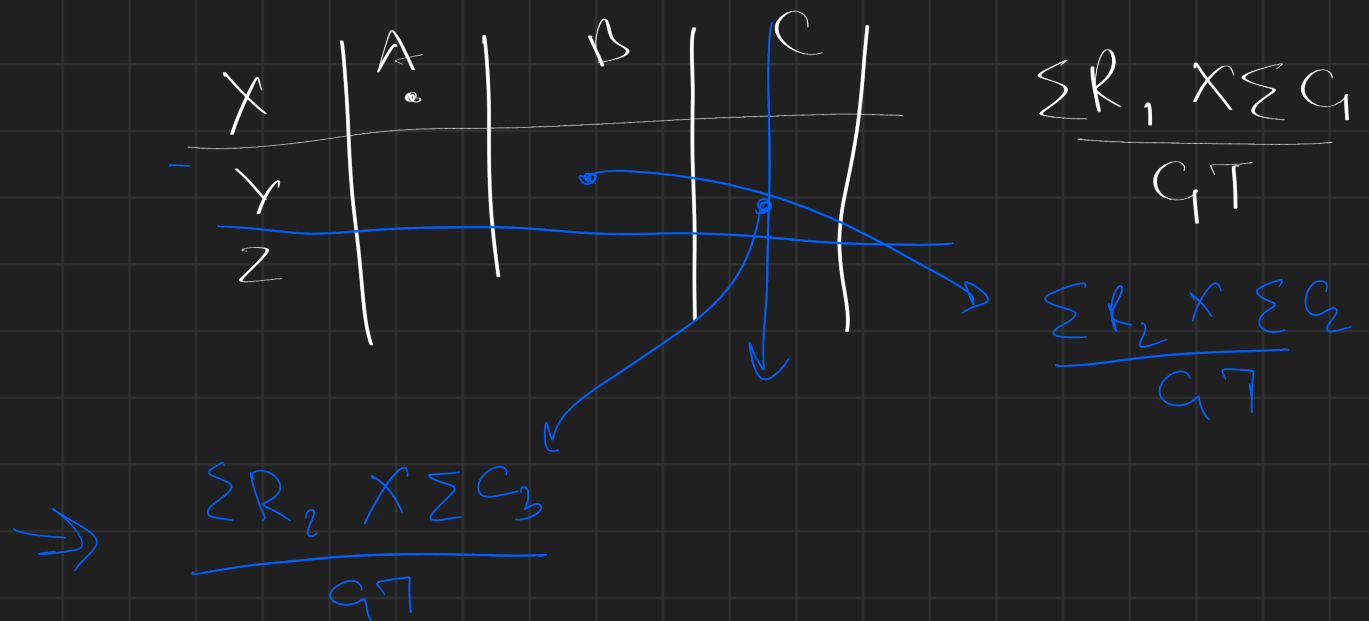
Cities	Products	Contingency Table	$\downarrow$ Cities	A	B	C
X	A		X			
Y	A		Y			
Y	B		Z			
Z	C					
Y	C					
Z	C					

	A	B	C	
X	O_{ij}			$\rightarrow \sum R_1$
X				$\rightarrow \sum R_2$
Z				$\rightarrow \sum R_3$
	$\sum C_1$	$\sum C_2$	$\sum C_3$	$\rightarrow C_1^T$

Column Total

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

$$E_{ij}^o = \frac{\sum R_i \times \sum C_j}{G_T}$$



$$DoF = ? \Rightarrow (\# \text{ Rows} - 1) \times (\# \text{ Col} - 1)$$

$$\begin{cases} \bullet & p\text{-value} < \alpha \rightarrow \text{Reject } H_0 \\ \bullet & \chi^2_{\text{Tab}} < \chi^2_{\text{computed}} \rightarrow \text{Reject } H_0 \end{cases}$$

2) Bayes Theorem:

$$P(A) = \frac{\text{card}(A)}{\text{card}(U)}$$

$P(A \cap B) \rightarrow A \text{ and } B \text{ Joint}$

$$P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$$

$P(X|Y) \rightarrow \text{Conditional}$

$P(A) \rightarrow \text{Marginal}$

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Posterior ← Likelihood ← Prior ← Marginal

$$\left\{ P(A|B) = \frac{P(B|A) \cdot P(A)}{\sum P(B|A_i) \cdot P(A_i)} \right\}$$

$$100 \rightarrow \left(\begin{array}{c} 80 \text{ H-F} \\ 5 \text{ False Pos} \end{array} \right) \} \frac{80}{85} \rightarrow 93\%$$

$$100 \rightarrow \left(\begin{array}{c} 1 \text{ No Dis} \\ 5 \text{ False Pos} \end{array} \right) \} \frac{1}{6} \rightarrow \left(\frac{2}{6} \right)$$

180
1000

PYQs

(1)

You want to apply for a student visa to country X. You can do this through any one of the two application centers, "A" or "B". To determine which application center to choose, you collect data. Currently, the embassy has decided to receive 65% of the applications from center A, and 35% from center B. Historically, 70% of the applications from center A have been granted the visa, and 80% of the applications from center B have been granted the visa. Then answer the following questions

(a)

What is your probability of applying for a student visa through center A and it is granted?

(0.455)

(b)

What is your probability of applying for a student visa through center B and it is granted?

(c)

Say you find someone who had applied for a student visa in the past, and the application was rejected. However, that person (someone) has not told you which application center was chosen to submit the application. Then which application center will you choose, given this additional information?



$$P(? | \sim G) \left| \begin{array}{l} P(B | \sim G) \\ < P(A | \sim G) \end{array} \right.$$

compare

$$P(A | \sim G)$$

$$P(B | \sim G)$$

~~0.195~~

$$P(A | \sim G) = \frac{P(A \cap \sim G)}{P(\sim G)} = \frac{0.65 \times 0.3}{P(\sim G)}$$

$$P(B | \sim G) = \frac{P(B \cap \sim G)}{P(\sim G)} = \frac{0.35 \times 0.2}{P(\sim G)}$$

2)

Four teams (1, 2, 3 and 4) have qualified for the final of the cricket world cup. Team 1 plays Team 2. Team 3 plays Team 4. The winners then play in the final. In a game between Team “i” and Team “j”, the probability of team “i” winning is given by $\{i / (i+j)\}$. Then what is the conditional probability that “Team 1” will win the World Cup?



$$P(1 \text{ wins}) = P(1-M_1)P(3-M_2)P(1-M_F) + P(1-M_1)P(4-M_2)P(1-M_F)$$

$$\frac{1}{3} \cdot \frac{3}{7} \cdot \frac{1}{4} + \frac{1}{3} \cdot \frac{4}{7} \cdot \frac{1}{5} = \frac{1}{20} + \frac{4}{105} = 0.0738$$

3)

A cosmetics brand “SwamilsBeautiful” manufactures three products, “Face Creams (FC)”, “Body Lotion (BL)” and “Lip Gloze (LG)”. The company believes displaying all their products together in a single shelf at a retailer store will help boost their sales, as people tend to pick up the items together. Hence, they are currently spending a lot of money to make sure that the retailers display the products together.

However, on hearing this, Dr.Milo feels sceptical as it is counter intuitive. He feels that the sales of the products are independent. To check his theory, data in Table-3 was collected. Table-3 contains the total sales (in quantity) of the three products across retail stores. Here, the retail stores were classified in two types: “Type-1: The three products are displayed together”, “Type-2: The three products are not displayed together”. Given this information answer the given subquestions.

(Note: For all questions in this comprehension, if your answer is in decimal, enter it rounded to

Product	Retail Store Type	
	Type-1	Type-2
FC	288	124
BL	622	204
LG	212	128

Table-3

(a)

What type of representation is Table-3?

- A) Contingency Table
- B) Frequency Table
- C) None of these
- D) Both Contingency Table & Frequency Table

What is the expected frequency for "BL" in Type-2 stores?

238.6920

$$\frac{456 \times 826}{1578}$$

What is the value of the computed test statistic for the hypothesis test that DrMilo will perform to check if his intuition is correct?

	①	②
FC	292.94	119.057
BL	587.90	238.69
LG	241.74	98.250

$$\sum \chi = \frac{\sum (O - E)^2}{E}$$

$$0.0833 + 0.2018 + 2.0502 + 5.04166 + 3.658 + 9.0082$$

What is the Null Hypothesis for the hypothesis test that DrMilo will be performing? (choose all that is applicable).

- The total sales is independent among the different products and different retail store types
- The total sales is not independent among the different products and different retail store types
- The total sales is uniformly distributed among the different products and different retail store types
- The total sales is uniformly distributed only among the different products . The total sales is uniformly distributed only among the different retail store types

$$\chi^2_{\text{Tab}}(\alpha = 0.2, \text{Dof} = 2) < \chi^2_{\text{Obs}}$$

Given Figure-2, at a 20% level of significance, what will be the conclusion of the hypothesis test? (choose all that is applicable)

$$\alpha = 0.2$$

- Reject the Null Hypothesis

- Do Not Reject the Null Hypothesis

- Reject the Alternative Hypothesis

- Do Not Reject the Alternative Hypothesis

- Accept the Null Hypothesis

- Do Not Accept the Null Hypothesis

- Accept the Alternative

$$\text{Dof} = (3-1)(2-1) = 2 \times 1 = 2$$

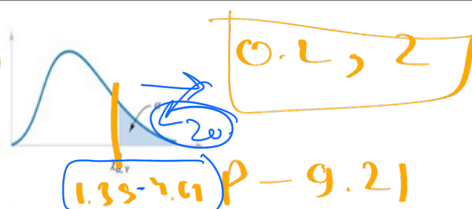


Table III Percentage Points $\chi^2_{\alpha, v}$ of the Chi-Squared Distribution

α	.995	.990	.975	.950	.900	.800	.700	.600	.500	.400	.300	.200	.100	.050	.025	.010	.005
1	.004	.004	.005	.006	.008	.010	.013	.016	.020	.025	.031	.038	.046	.054	.064	.075	.088
2	.010	.009	.011	.013	.016	.020	.025	.031	.038	.046	.054	.064	.075	.088	.101	.115	.130
3	.016	.015	.017	.019	.023	.028	.034	.041	.049	.058	.067	.077	.088	.100	.113	.127	.142
4	.020	.019	.021	.023	.027	.033	.039	.046	.054	.063	.072	.082	.093	.105	.119	.133	.148
5	.023	.022	.024	.026	.030	.036	.042	.050	.058	.067	.076	.086	.097	.109	.123	.137	.152
6	.025	.024	.026	.028	.032	.038	.044	.052	.060	.069	.078	.088	.099	.111	.125	.139	.154
7	.027	.026	.027	.029	.033	.039	.045	.053	.061	.070	.079	.089	.100	.112	.126	.140	.155
8	.028	.027	.028	.030	.034	.040	.046	.054	.062	.071	.080	.090	.101	.113	.127	.141	.156
9	.029	.028	.029	.031	.035	.041	.047	.055	.063	.072	.081	.091	.102	.114	.128	.142	.157
10	.030	.029	.030	.032	.036	.042	.048	.056	.064	.073	.082	.092	.103	.115	.129	.143	.158

Figure-2

(5)

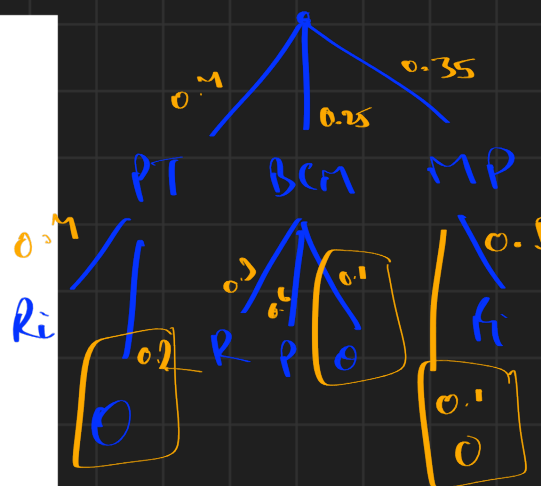
(The following comprehension is purely imaginary)

A restaurant ("Milo's Kitchen") has data pertaining to customers ordering three types of starters "Paneer Tikka (PT)", "Baby Corn Manchurian (BCM)" and "Masala Papad (MP)". For simplicity assume that the orders for starters is mutually exclusive and that a customer does not order more than one starter. Moreover, the customers who order these starters are predominantly one of the three types of eaters, "Roti Eaters", "Poori Eaters" or "Rice Eaters" (Note: customers can be "other eaters" as well. Hence, the list is not exhaustive). From past footfall data, it is seen that 25% of the customer's order "BCM" and another 40% order "PT".

The remaining customers order "MP". It is observed that,

- Of the people ordering "BCM", 30% are "Roti Eaters" and 60% are "Poori Eaters".
- For "PT" orders, it is observed that 40% are "Rice Eaters", 20% are "Poori Eaters" and another 20% are "Roti Eaters".
- 50% of the customers who order "MP" are "Rice Eaters" and 40% are "Poori Eaters".
- "BCM" is never ordered by "Rice Eaters" and "MP" is never opted for by "Roti Eaters".

Assuming any other necessary mutual exclusiveness, and answer the given sub questions.



(a)

If 1,000 customers come to the restaurant, how many will/may be "Rice Eaters"? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "10.235", then enter it as "10.24". DO NOT ENTER PERCENTAGE VALUES)

→ 335

$$\begin{aligned}
 P(\text{Rice}) &= P(\text{PT})P(\text{Ri}|\text{PT}) + P(\text{MP})P(\text{Ri}|\text{MP}) \\
 &= 0.4 \times 0.4 + 0.35 \times 0.5 \\
 &= 0.335 \text{ or } 335 \text{ out of } 1000 \text{ people}
 \end{aligned}$$

(b)

If 4,000 customers had visited the shop on 10th December 2024, then how many "BCMs" will you expect to serve? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "10.235", then enter it as "10.24". DO NOT ENTER PERCENTAGE VALUES)

$$4000 \times 0.25 = 1000$$

(c)

Milo's Kitchen is planning spend money on purchasing a refrigerator (for "Paneer" and "Baby Corn" storage). If the 600 "Poori Eaters" are expected to arrive, then what is the expected revenue generated from them if the restaurant charges Rs. 350 per PT order, Rs. 250 per BCM order and Rs. 80 per MP order? (Note: Enter the answer "in Rupees" rounded to two decimal places. For example, if the answer is "Rs. 10.235", then enter it as "10.24". DO NOT ENTER ANY SYMBOLS OR LETTERS)

$$P(? | \text{Poori}) = \frac{P(\text{Poori} | ?) P(?)}{P(\text{Poori})}$$

[? : MP, BCM, PT]

$$\begin{aligned}
 P(\text{Poori}) &= P(\text{PT})P(\text{Poori}|\text{PT}) + P(\text{BCM})P(\text{Poori}|\text{BCM}) \\
 &\quad + P(\text{MP})P(\text{Poori}|\text{MP})
 \end{aligned}$$

$$= 0.4 \times 0.2 + 0.25 \times 0.6 + 0.35 \times 0.7$$

$$= 0.37$$

$$P(\text{BCM} | \text{Poor}) = \frac{0.15}{0.37} = 0.4054$$

$$P(\text{PT} | \text{Poor}) = \frac{0.02}{0.37} = 0.2162$$

$$P(\text{MP} | \text{Poor}) = \frac{0.14}{0.37} = 0.3783$$

Number

243.24

129.72

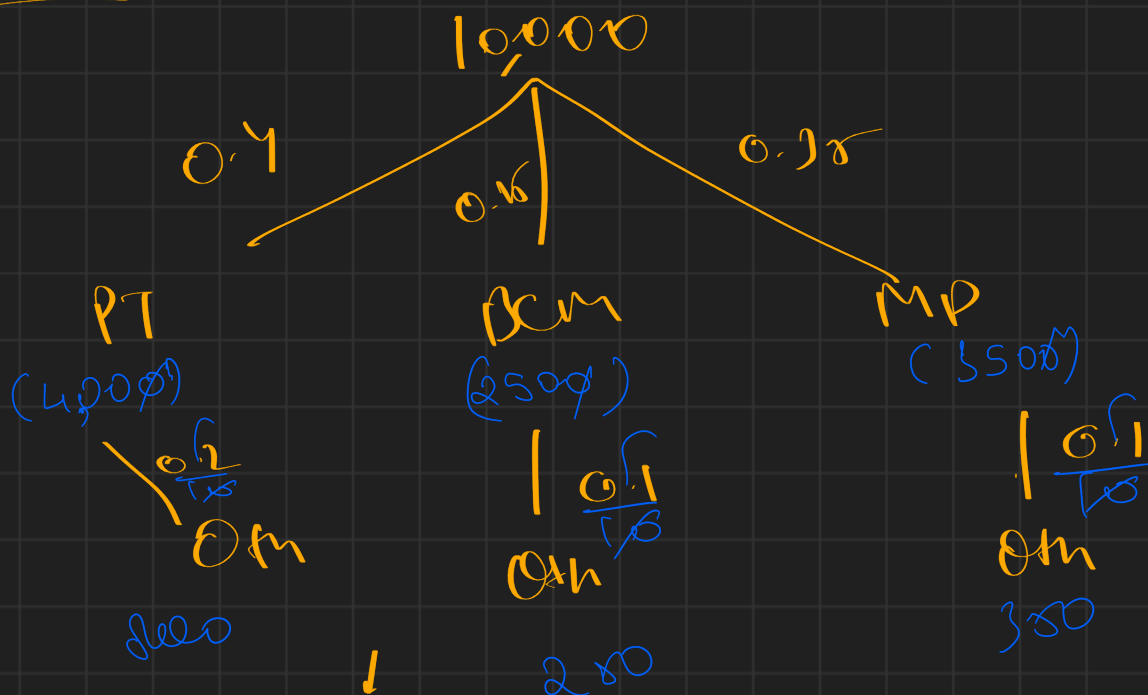
226.98

$$243.24 \times 250$$

$$+ 129.72 \times 330$$

$$+ 226.98 \times 80$$

$$= 124370.4$$



(d)

If a total of 10,000 customers visit Milo's Kitchen in a month, how many (in Total) will you expect to be non-Rice Eater, Non-Poori Eater and Non-Roti Eater? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "10.235", then enter it as "10.24". DO NOT ENTER PERCENTAGE VALUES)

1400 (Others)

(e) $P(I|om) \propto P(om|I)P(I)$

Recently, the owner found CCTV footage of a customer stealing a starter (but was not able to see which starter was stolen). The owner was sure that the food-stealing customer was not a "Rice Eater", "Poori Eater" or "Roti Eater". Then which starter would the owner suspect the stealing customer to have stolen?

- PT
- BCM
- MP
- None of these as Milo's Kitchen owner does not have a working CCTV (P.S: This is not an answer so

$P(PT|om) \propto P(om|PT)P(PT)$

$P(BCM|om) \propto P(om|BCM)P(BCM)$

$P(MP|om) \propto P(om|MP)P(MP)$

$\rightarrow 0.2 \times 0.4 = 0.08$

$\rightarrow 0.1 \times 0.25 = 0.025$

$\rightarrow 0.1 \times 0.25 = 0.025$